Computational strengthenings of Clebsch syndrome rigidity

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Abstract

For a projective arc $A \subset \mathrm{PG}(2,q)$, let $\mathcal{U}(A)$ be the points on no chord of A. The geometric paper proves, without an exhaustive classification of six-arcs over $\mathbb{F}_{11}$, that a six-arc in $\mathrm{PG}(2,11)$ whose uncovered locus lies on a conic is the Clebsch hexagon. Here exact finite computation sharpens and extends that result. There are fifteen projective classes of six-arcs over $\mathbb{F}_{11}$; the Clebsch class is the unique one whose uncovered locus is contained in a cubic, and it is separated from every other class by a four-point gap in uncovered-set size. A Sylvester-graph obstruction shows that $q = 11$ is the only field order admitting a conic-filling six-arc. Exhaustive orbit searches then classify all conic-filling arcs through eight points: only the projective four-frame over $\mathbb{F}_5$ and the Clebsch six-arc over $\mathbb{F}_{11}$ occur. The companion also preserves the original $q = 13$ computations underlying forthcoming Paper IV, which gives the passant-code reconstruction theorem a standalone structural and reproducible account [10]. The finite claims are accompanied by exact replay routes and a claim-by-claim trust ledger.

Keywords. finite projective plane; arc; conic; MDS code; deep hole; exhaustive classification.

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1 Setup and relation to the geometric paper

A *k-arc* is a set A of k points of $\mathrm{PG}(2,q)$ with no three collinear. Its chords are the joins of pairs of vertices, and

$$\mathcal{U}(A) = \{P \in \mathrm{PG}(2,q) : P \text{ lies on no chord of } A\}.$$

We call A *conic-filling* when $\mathcal{U}(A)$ is the full rational point set of a nonsingular conic. If n_i counts the off-arc points incident with exactly i chords, put

$$r = \lfloor k/2 \rfloor, \qquad t(A) = \sum_{i=3}^{r} \binom{i-1}{2} n_i, \qquad m = \binom{k}{2}.$$

The geometric paper [9] proves the following three inputs. They are restated to make every dependency in the finite arguments explicit.

Theorem 1.1 (Chord defect [9, Theorem 3.1])**.** *For every k-arc with $k \geq 4$,*

$$|\mathcal{U}(A)| = q^2 - (m-1)q + \left(m + 3\binom{k}{4} - k + 1\right) - t(A), \qquad 0 \leq t(A) \leq 3\binom{k}{4}\frac{r-2}{r}.$$

Theorem 1.2 (Conic-filling window [9, Theorem 4.4])**.** *If $\mathcal{U}(A)$ is a nonsingular conic, then q is odd and*

$$2k-3 \le q \le \frac{k(k-1)+3}{3} < \binom{k}{2}.$$

For $k=6$, writing $c(A)=n_3$ gives

$$|\mathcal{U}(A)| = q^2 - 14q + 55 - c(A), \qquad c(A) = (q-6)(q-9)$$

when $|\mathcal{U}(A)| = q+1$, and for q odd the concurrence spectrum restricts the count to $c(A) \in \{0,1,2,3,4,6,10\}$ [9, Proposition 3.2].

Theorem 1.3 (Geometric rigidity [9, Theorem 1.1])**.** *For a six-arc $A \subset \mathrm{PG}(2,11)$, the locus $\mathcal{U}(A)$ lies on a conic if and only if A is projectively equivalent to the Clebsch hexagon. In that case $\mathcal{U}(A)$ is exactly a nonsingular conic.*

The columns of a parity-check matrix of a projective $[k, k-3, 4]_q$ MDS code form a k-arc. When $\mathcal{U}(A)$ is nonempty it is the projective distance-three syndrome locus. Projective equivalence of the unordered column sets is the same as monomial equivalence of the codes. Thus each geometric classification below has the stated coding interpretation.

2 The six-arc census over $\mathbb{F}_{11}$

Computer-assisted conic-inscribed subcensus. For comparison, the exact replay enumerates all 1548 frame-normalized six-arc presentations and selects the 252 whose vertices lie on a nonsingular conic. Their $|\mathcal{U}(A)|$ histogram is

$\lvert\mathcal{U}(A)\rvert$	18	19	20	22
frequency	30	60	90	72

so $|\mathcal{U}(A)| \in \{18,19,20,22\}$. The replay is `check_rigidity_degenerate_conic.py`.

Table 1 records the complete census needed below. Every representative contains the standard frame

$$(1,0,0),\ (0,1,0),\ (0,0,1),\ (1,1,1);$$

the table gives the remaining two points of each representative. Here $d_{\min}$ is the least degree of a nonzero form vanishing on the uncovered locus.

class	pair	$\lvert G\rvert$	$\lvert\mathcal{U}\rvert$	$d_{\min}$
C01	23/32	2	20	5
C02	23/34	6	18	4
C03	23/36	3	19	5
C04	23/37	3	19	5
C05	23/39	4	20	5
C06	23/42	2	20	5
C07	23/48	1	21	5
C08	23/49	4	20	5
C09	23/56	4	20	5
C10	23/62	6	19	5
C11	23/67	12	16	4
C12	23/74	5	22	6
C13	23/89	6	18	4
C14	23/(9,10)	12	18	4
C15	34/45	60	12	2

Table 1: The fifteen projective classes of six-arcs in $\mathrm{PG}(2,11)$. In the pair column, xy/uv abbreviates $(1,x,y),(1,u,v)$. Class C15 is the Clebsch class. Both replay programs regenerate the same canonical keys: `check_global_conic_gap.py` supplies $|G|$ and $|\mathcal{U}|$, and `check_low_degree_loci.py` supplies $d_{\min}$.

The direct certificate records, for each row, a canonical representative, its stabilizer, its orbit mass, its number $c(A)$ of triple chord concurrences, and its conic-intersection histogram. It checks

$$|G_A| m_A = 360, \qquad \sum_A m_A = 1548, \qquad |\mathcal{U}(A)| = 22 - c(A).$$

Thus the fifteen rows and every uncovered-set size follow from the orbit ledger and the chord-defect identity. The same counts corroborate the concurrence spectrum of the geometric paper [9, Proposition 3.2]: across the fifteen classes $c(A) = 22 - |\mathcal{U}(A)|$ takes each of the seven values $0, 1, 2, 3, 4, 6, 10$ and none of the excluded values five, seven, eight, and nine. This is corroboration at $q = 11$, not a second proof of the proposition. Regenerating all 1548 normalized arcs remains an independent audit rather than the reader-facing classification proof.

The same census supports a strengthening in which a cubic, rather than a conic, is allowed.

Proposition 2.1 (Low-degree algebraic rigidity)**.** *For a six-arc $A \subset \mathrm{PG}(2, 11)$, the uncovered locus $\mathcal{U}(A)$ is contained in the zero set of a nonzero homogeneous form of degree at most three if and only if A is projectively equivalent to the Clebsch hexagon. In the Clebsch case the least such degree is two: the quadratic vanishing space is spanned by the defining nonsingular conic equation Q, and the cubic vanishing space is*

$$Q \cdot \mathbb{F}_{11}[X, Y, Z]_1.$$

Proof by finite certificate. For each of the fourteen non-Clebsch orbit representatives, the certificate selects ten points of $\mathcal{U}(A)$ and gives a nonsingular 10-by-10 evaluation minor for the cubic monomials. Hence the full cubic evaluation map has rank ten. A containing form of smaller positive degree could be multiplied by a nonzero monomial to produce such a cubic, so smaller positive degrees are excluded as well; no nonzero constant vanishes on the nonempty locus. For the Clebsch representative, direct symbolic reduction proves that the quadratic kernel is generated by Q and that the cubic kernel is generated by QX, QY, QZ. The orbit-mass identities above show that these fifteen local witnesses cover every projective class. □

Corollary 2.2 (Monomial characterization)**.** *Up to monomial equivalence, there is a unique linear $[6, 3, 4]_{11}$ code of covering radius three whose projective deep-hole syndrome locus is contained in a plane curve of degree at most three. It is the Clebsch code, whose projective stabilizer is A_5.*

Proof. The columns of a parity-check matrix of such a code form a six-arc A, and covering radius three identifies its projective deep-hole syndrome locus with $\mathcal{U}(A)$. Proposition 2.1 makes A projectively equivalent to the Clebsch hexagon. A projectivity between two unordered parity-check column sets lifts, after choosing column representatives, to a row operation together with a coordinate permutation and nonzero coordinate scalings; these are exactly a monomial code equivalence. Conversely, the Clebsch code has the stated property, and Dye's Theorem 3 gives the A_5 statement [5, p. 278]. □

Remark 2.3 (Sharpness of the degree threshold). The cutoff cannot be raised from three to four: the four non-Clebsch classes `C02`, `C11`, `C13`, and `C14` have $d_{\min} = 4$, so the failure is not isolated. For `C02`, `check_low_degree_loci.py` prints a quartic whose rational zero set is exactly the 18-point uncovered locus and verifies the equality point by point. No smoothness, irreducibility, or genus assertion is used here.

Theorem 2.4 (Numerical gap)**.** *The Clebsch class has $|\mathcal{U}(A)| = 12$, while every non-Clebsch six-arc has $|\mathcal{U}(A)| \geq 16$. Equivalently, $|\mathcal{U}(A)| \leq 15$ characterizes the Clebsch class, for which $|\mathcal{U}(A)| = 12$. If $|\mathcal{U}(A)| \geq 16$, then $\mathcal{U}(A)$ lies on no conic.*

Proof by finite certificate and exhaustive audit. Any ordered four points of a six-arc form a projective frame. The certificate gives fifteen canonical representatives and checks their stabilizers and orbit masses, whose sum is 1548. Since every normalized six-arc has a canonical key and the full-domain audit finds exactly those keys, the ledger is complete. The chord-defect formula then derives $|\mathcal{U}(A)|$ from the certified concurrence count in each row. Their extension counts are

$$|\mathcal{U}(A)| \in \{12, 16, 18, 19, 20, 21, 22\},$$

with multiplicities

$$(6, 30, 150, 300, 630, 360, 72)$$

across the 1548 normalized representatives. The six presentations attaining 12 form the single Clebsch class; every other class has at least 16 extension points. Theorem 1.3 supplies the final conic-containment assertion, and the four-point gap between twelve and sixteen is the sharpness figure it leaves. The orbit ledger also records, for each class, the largest intersection achieved by any of the 160,930 nonsingular conics of the plane, together with one conic attaining it. Rigidity already excludes conic containment for every non-Clebsch class, so that enumeration is reported computation rather than a premise of the theorem. □

3 Cross-field uniqueness for six-arcs

Lemma 3.1 ($q = 9$ polarity graph)**.** *For a nonsingular conic $\mathcal{C} \subset \mathrm{PG}(2, 9)$, let Σ be the graph on the* 36 *internal points in which*

$$P \sim Q \quad \iff \quad Q \in P^{\perp}$$

for the conic polarity. Then Σ is the Sylvester graph, with intersection array

$$\{5, 4, 2; 1, 1, 4\}.$$

For distinct internal points P, Q, the line PQ is passant to $\mathcal{C}$ if and only if P and Q are at distance two in Σ, and the graph joining pairs at distance two has clique number

$$\omega(\Sigma_2) = 5.$$

Proof. Counting internal points on polar lines and their intersections gives $b_0 = 5$, $b_1 = 4$, $b_2 = 2$, $c_1 = c_2 = 1$, and $c_3 = 4$, hence the displayed intersection array. This is the Sylvester graph [4, §13.1.2]; uniqueness for this intersection array is proved in [7, Thm. 6].

For distinct internal points P, Q, put $R = P^{\perp} \cap Q^{\perp} = (PQ)^{\perp}$. If PQ is passant, then its pole R is internal and adjacent to both P and Q; since the intersection array has $a_1 = 0$, these vertices are at distance two. Conversely, a common neighbor of P and Q must equal R, so distance two makes the pole of PQ internal and hence makes PQ passant. The final value is $\mathrm{eq}_2(\Sigma) = 5$ in [1, Table 1]. □

Theorem 3.2 (Uniqueness of $q = 11$)**.** *Let q be a prime power. If a six-arc $A \subset \mathrm{PG}(2, q)$ satisfies $\mathcal{U}(A) = \mathcal{C}(\mathbb{F}_q)$ for some nonsingular conic $\mathcal{C}$, then $q = 11$. At $q = 11$ every such A is projectively equivalent to the Clebsch hexagon.*

Proof. Theorem 1.2 gives $q \geq 9$, $c(A) = (q-6)(q-9)$, and $c(A) \leq 10$ from the concurrence spectrum. Thus the only prime-power candidates are $q = 9, 11$: $q = 10$ is not a prime power, and every $q \geq 12$ gives $c(A) \geq 18$.

It remains to exclude $q = 9$. Since every point of $\mathcal{C}$ is uncovered, no chord of A meets $\mathcal{C}$; every chord is therefore passant. A point off $\mathcal{C}$ lies on five passants if it is internal and four if it is external, so the five chords through each vertex force all six vertices of A to be internal. Lemma 3.1 would then make A a six-clique in the distance-two graph of the Sylvester graph, whose clique number is five. This is impossible.

Therefore $q = 11$. The final assertion follows from Theorem 1.3, which puts every conic-filling six-arc over $\mathbb{F}_{11}$ in the single Clebsch orbit. $\square$

Remark 3.3 (Why characteristics three and five need separate care). The golden normal form of the geometric paper evaluates the associated polarity on the configuration exactly over $\mathbf{Z}[\varphi]$: the six vertices give values of norm -5 and the ten Brianchon points values of norm -9 [9]. So the vertices miss the associated conic unless the characteristic is five, and the Brianchon points miss it unless the characteristic is three. Neither exception is vacuous. In characteristic three the configuration exists, since $5 = -1$ is a square in $\mathbb{F}_9$, but all ten of its Brianchon points lie on its associated conic; the $q = 9$ case above therefore cannot be closed by conic avoidance, and it is the Sylvester clique bound that closes it. In characteristic five the six vertices themselves lie on the associated conic and the two golden roots coincide, which is why the $q = 5$ entry of Theorem 5.1 is a four-frame rather than a six-arc.

4 The binary passant/internal incidence code at $q = 13$

Fix the conic $\mathcal{C} : XZ - Y^2 = 0$ in $\mathrm{PG}(2, 13)$. Let M be the 78-by-78 incidence matrix over $\mathbb{F}_2$, with rows indexed by the passant lines to $\mathcal{C}$ and columns indexed by its internal points. Madison and Wu determine the kernel of this matrix for every odd q. Over an algebraic closure of $\mathbb{F}_2$ they give both its dimension $(q-1)^2/4$, which is 36 at $q = 13$, and its exact decomposition into pairwise nonisomorphic simple modules for $\mathrm{PSL}(2, q)$ [8, Theorem 6.1, Corollary 6.3]. A descent refinement of that decomposition is recorded in the project's research notes: over $\mathbb{F}_2$ itself, and for the full $\mathrm{PGL}(2, 13)$-action, $K = \ker M$ is irreducible with endomorphism field $\mathbb{F}_8$; equivalently K carries a canonical twelve-dimensional $\mathbb{F}_8$-structure whose base change to the closure returns Madison and Wu's three summands as Galois conjugates. Its proof is not reproduced here and its formalization is pending, so it is outside the claim surface of this artifact and nothing below rests on it: the spanning conclusion of Theorem 4.2 is proved from the mod-two intersection algebra instead. Hollmann and Xiang construct the elliptic association scheme afforded by the $\mathrm{PGL}(2, q)$-action on the passant carrier [6, §3]. Thus the dimension and the association scheme used below are known. The assertions here are the exact minimum distance and the intrinsic reconstruction of the incidence matrix and scheme from the minimum words. They are retained as the historical computational source. Their current paper-level development belongs to forthcoming Paper IV [10], which supplies a standalone proof and evidence surface.

One numerical coincidence drives both the weight-eight exclusion below and the terminal $k = 8$ argument of Section 5. We isolate it.

Lemma 4.1 (Pencil saturation at $q = 2k - 3$)**.** *Let $\mathcal{C}$ be a nonsingular conic in $\mathrm{PG}(2, q)$ with q odd, and let A be a k-arc of internal points all of whose joins are passant to $\mathcal{C}$. If $q = 2k - 3$, then the $k-1$ joins at a vertex P exhaust the passant pencil at P. The remaining $q + 2 - k$ lines through P are then exactly the conic-secants through P, and they are exactly the tangent lines of A at P.*

Proof. An internal point lies on exactly $(q + 1)/2$ passants and on no tangent. The $k - 1$ joins at P are distinct because A is an arc and passant by hypothesis, and $k - 1 = (q + 1)/2$ is precisely the condition $q = 2k - 3$; so they are the entire passant pencil at P. The other $q + 2 - k$ lines through P are therefore non-passant, hence secant. Each meets A only in P, because every other point of A is joined to P by a passant, so each is a tangent of A at P; and A has exactly $q+2-k$ tangents at each vertex. $\square$

Theorem 4.2 (Passant-code theorem). *The binary code $K = \ker M$ has parameters*

$$[78, 36, 12]_2.$$

There are exactly 364 minimum words. Under $\mathrm{PGL}(2, 13)$ they form four orbits of size 91, one with stabilizer S_4 and three with stabilizer D_{24}. Each orbit spans K.

The minimum supports intrinsically reconstruct the full six-class elliptic association scheme on the 78 internal points. Pair concurrence recovers passant versus secant join type, triple concurrence separates the six orbitals, and the 78 all-zero-triple seven-cliques are exactly the passant incidence rows. Consequently the minimum layer reconstructs M up to row order, and its automorphism group is $\mathrm{PGL}(2, 13)$.

Proof. Every row and column of M has weight seven, and conic polarity identifies the two index sets. Madison and Wu's dimension formula gives $\dim K = 36$, equivalently $\operatorname{rank} M = 42$; exact elimination independently checks this specialization. Summing the row equations gives the all-one functional, so every word of K has even weight. Moreover, if a nonzero word contains an internal point P, each of the seven passants through P must contain another support point. These seven points are distinct, proving the elementary lower bound $\mathrm{wt} \geq 8$.

Suppose first that a word has weight eight. The preceding parity argument shows that its support is an eight-arc of internal points, every join of two of which is passant. Since $13 = 2 \cdot 8 - 3$, Lemma 4.1 applies: the seven remaining lines through each vertex are simultaneously the seven conic-secants through that internal point and the seven tangents of the arc there.

For an internal point P, let

$$T_P(X) = \prod_{\substack{\ell \ni P \\ \ell \text{ secant to } \mathcal{C}}} \ell(X).$$

For a pairwise-passant triple put

$$h(P, Q, R) = \frac{T_P(Q) T_Q(R) T_R(P)}{T_P(R) T_R(Q) T_Q(P)}.$$

Segre's lemma of tangents [2, Lemma 27] gives $h(P, Q, R) = 1$ for every triple in the hypothetical support. Fixing $P = (1 : 0 : 2)$ therefore makes its other seven points a clique in the graph on the 42 passant-join neighbors of P, where Q and R are adjacent when QR is passant and $h(P, Q, R) = 1$.

An order-14 element of the stabilizer of P splits these vertices into three cyclic orbits A_i, B_i, C_i, with indices in $\mathbf{Z}/14$. Direct substitution in the displayed product gives the complete adjacency rule

orbit pair	$j - i \pmod{14}$
AA	$4, 6, 8, 10$
AB	$6, 7, 11, 12$
AC	$1, 3$
BB	$6, 8$
BC	$3, 5, 6, 8, 9, 11$
CC	$2, 4, 10, 12.$

Up to simultaneous translation, every four-clique is one of five rows:

four-clique	unique extension
A_0, B_6, B_{12}, C_1	C_3
A_0, B_6, B_{12}, C_3	C_1
A_0, B_6, C_1, C_3	B_{12}
A_0, B_{12}, C_1, C_3	B_6
B_0, B_6, C_9, C_{11}	A_8.

They close to fourteen translates of one five-clique, none of which extends. The local clique number is therefore five, not seven. This excludes weight eight.

For weight ten, fix one support point. Parity leaves only two passant-pencil profiles. If s of the other nine support points are joined to the fixed point by secants, the remaining $9 - s$ points are distributed in positive odd numbers among its seven passants. Hence s is even and $9 - s \geq 7$, so $s = 0$ or 2. These give, respectively, three points on one passant and one on each of the other six, or one on every passant together with two secant-join neighbors. The two possibilities are excluded by compact syndrome certificates. Write m_Q for the 78-bit incidence column of an internal point Q. In the $s = 0$ profile, each choice of exceptional fibre gives two canonically sorted sets of partial XORs, of sizes 216 and 4320; in the $s = 2$ profile the corresponding sizes are 216 and 771024. The certificate checks that the two sets are disjoint in every case. These checks exhaust respectively

$$7\binom{6}{3}6^6 = 6{,}531{,}840, \qquad 6^7\binom{35}{2} = 166{,}561{,}920$$

supports, so neither profile has zero syndrome. An independent dynamic program builds all full-fibre XORs with a different split and obtains the same two empty intersections. Finally, the following twelve internal points do have zero syndrome:

$$\begin{gathered}(1:0:2), (1:3:2), (1:4:5), (1:1:8),\\ (1:4:8), (1:1:7), (1:7:12), (1:3:3),\\ (1:9:11), (1:10:11), (1:0:5), (1:8:7).\end{gathered}$$

Their stabilizer is dihedral of order 24. This proves $d(K) = 12$.

The classification of the minimum layer is an exhaustive exact execution. The replay generates four projective orbits from explicit representatives and verifies all 364 supports against M. It then forms all pair and triple concurrence profiles. Pair concurrences are $7, 9, 12$ on passant joins and $6, 8$ on secant joins; the six triple histograms are distinct. Exactly 78 of the resulting passant seven-cliques have zero triple concurrence, and direct comparison identifies them with the rows of M.

The span and automorphism conclusions have a shorter structural proof. Put

$$\Delta(x, y, z) = y^2 - xz, \qquad \beta(P, Q) = 2y_P y_Q - x_P z_Q - z_P x_Q, \qquad \rho(P, Q) = \frac{\beta(P, Q)^2}{\Delta(P)\Delta(Q)}.$$

On distinct internal points, ρ takes the six values $0, 1, 3, 9, 10, 12$. Its level relations are the six classes of the elliptic association scheme of Hollmann and Xiang [6, §3]; write A_r for the adjacency matrix of the relation $\rho = r$. Conic polarity identifies an internal point with its passant polar, so A_0 is M, up to row order.

For clarity, the part of the intersection algebra used here can also be checked directly. Fixing one point and substituting one representative of each ρ-relation gives, over $\mathbb{F}_2$,

$$\begin{aligned} A_0^2 &= I + A_9 + A_{10} + A_{12}, \\ A_0 A_r &= 0 \quad (r = 9, 10, 12), \\ A_9^2 &= A_{10}, \qquad A_{10}^2 = A_{12}, \qquad A_{12}^2 = A_9. \end{aligned} \tag{1}$$

These are parity reductions of ordinary integer intersection numbers, not identities inferred from matrix rank. For example the coefficients of A_0^2 on $I, A_0, A_1, A_3, A_9, A_{10}, A_{12}$ are $(7, 0, 0, 0, 1, 1, 1)$; those of $A_0A_9, A_0A_{10}, A_0A_{12}$ are respectively

$$(0, 2, 2, 2, 0, 2, 0), \quad (0, 2, 2, 0, 2, 0, 2), \quad (0, 2, 0, 2, 0, 2, 2),$$

and those of $A_9^2, A_{10}^2, A_{12}^2$ are respectively

$$(14, 0, 4, 2, 2, 1, 4), \quad (14, 0, 2, 4, 2, 4, 1), \quad (14, 4, 2, 2, 3, 2, 2).$$

Let $B = A_9$. Since $A_0B = 0$, the image of B lies in $K = \ker A_0$. If $x \in K$ and $Bx = 0$, the first identity in (1), together with $B^2 = A_{10}$ and $B^4 = A_{12}$, gives

$$0 = A_0^2 x = (I + B + B^2 + B^4)x = x.$$

Thus B is injective on the 36-dimensional space K, so $\operatorname{im} B = K$ and B, B^2, B^4 all have rank 36. If N_i is the 91-by-78 support matrix of the ith minimum-word orbit, direct pair counting in one representative orbit gives

$$(N_1^\mathsf{T} N_1, N_2^\mathsf{T} N_2, N_3^\mathsf{T} N_3, N_4^\mathsf{T} N_4) = (A_9, A_9, A_{12}, A_{10}).$$

Hence every N_i has rank at least 36; its rows lie in K, so every minimum-word orbit spans K.

Finally, the six concurrence colors recover exactly the six ρ-relations, so an automorphism of the minimum layer preserves the elliptic scheme. The following four internal points form a geometric base:

$$P_0 = (1 : 0 : 2), \quad P_1 = (1 : 1 : 7), \quad P_2 = (1 : 0 : 7), \quad P_3 = (1 : 1 : 3).$$

Their first three pair relations are $(\rho(P_0, P_1), \rho(P_0, P_2), \rho(P_1, P_2)) = (10, 3, 9)$. There are $78 \cdot 14 \cdot 2 = 2184$ ordered triples with this relation pattern: the relation A_{10} has valency 14, and the relevant intersection number is $p_{3,9}^{10} = 2$. Direct substitution in the symmetric-square action shows that the stabilizer of (P_0, P_1, P_2) in $\mathrm{PGL}(2, 13)$ is trivial. Since that group also has order 2184, it acts simply transitively on these triples. For a fixed such triple, P_3 is the unique internal point with relation signature $(3, 1, 9)$, and the four values

$$(\rho(P, P_0), \rho(P, P_1), \rho(P, P_2), \rho(P, P_3)),$$

with equality to an anchor recorded on the diagonal, distinguish all 78 internal points. Both assertions follow by substitution in the displayed formula for ρ; the exact replay checks all 78 normalized internal points independently. Therefore a scheme automorphism can first be composed with a unique element of $\mathrm{PGL}(2, 13)$ so that it fixes the first three anchors; it then fixes the fourth anchor and hence every internal point. Thus the reconstructed minimum layer has automorphism group exactly $\mathrm{PGL}(2, 13)$. The original rank eliminations and the independent order-2184 stabilizer-chain enumeration remain in the replay as checks. □

5 The small-arc boundary

The chord moments reduce the neighboring cases through eight points to a finite list of fields. Exact terminal enumeration excludes the remaining seven- and eight-point cases.

Theorem 5.1 (Conic-filling loci through eight points). *Let A be a k-arc in $\mathrm{PG}(2,q)$, where q is a prime power and $4 \le k \le 8$, and suppose that $\mathcal{U}(A)$ is the full set of $\mathbb{F}_q$-rational points of a nonsingular conic. Then*

$$(k,q) = (4,5) \qquad \textit{or} \qquad (k,q) = (6,11).$$

In the first case A is a projective four-frame; in the second it is a Clebsch hexagon. Conversely, both cases occur.

Moreover, for each $q \in \{13, 17, 19\}$, an arc all of whose chords are passant to a fixed nonsingular conic has at most six points, and this bound is attained.

Proof with exhaustive finite verification. For $k = 4$, Theorem 1.2 gives $5 \le q < 6$, hence $q = 5$. Every four-arc is a projective frame, and for the standard frame a direct substitution gives

$$\mathcal{U}(A) = V(X^2 + Y^2 + Z^2 + XY + XZ + YZ),$$

where $V(f)$ denotes the projective zero set of f; this is a nonsingular six-point conic. For $k = 5$, Theorem 1.1 gives $|\mathcal{U}(A)| = q^2 - 9q + 21$; equality with $q+1$ would require $q^2 - 10q + 20 = 0$, which has no integral root. The case $k = 6$ is Theorem 3.2 together with Theorem 1.3.

For $k = 7$, Theorem 1.1 gives

$$|\mathcal{U}(A)| = q^2 - 20q + 120 - n_3.$$

If this equals $q + 1$, then

$$n_3 = q^2 - 21q + 119, \quad n_2 = 105 - 3n_3, \quad n_1 = 3(q^2 - 14q + 42).$$

The strengthened conic-filling window gives $11 \le q \le 15$. Thus only the prime powers $q = 11, 13$ remain, with forced spectra $(n_1, n_2, n_3) = (27, 78, 9)$ and $(87, 60, 15)$ respectively. An exhaustive four-frame-normalized audit finds 10232 distinct normalized seven-arcs at $q = 11$, of which 140 have $|\mathcal{U}(A)| = q+1$, and 53960 at $q = 13$, of which 1680 have that size. The compact certificate quotients these 1820 candidates into one projective orbit at $q = 11$ and two at $q = 13$, with respective masses $140, 840, 840$. A nonsingular 6-by-6 quadratic evaluation minor for each orbit excludes conic containment. The three mass identities and the normalized-domain audit prove completeness. Every seven-arc is reached because it contains a four-frame; extending each normalized six-arc by its uncovered points produces every resulting seven-arc exactly three times, according to which non-frame point is deleted.

Finally let $k = 8$. The strengthened conic-filling window gives $13 \le q \le 59/3$, so the odd prime powers are exactly $13, 17, 19$.

It remains to exclude these three fields. By projective equivalence fix the conic $XZ = Y^2$, whose stabilizer is the symmetric-square image of $\mathrm{PGL}(2,q)$. Its complement has q^2 points. In the terminology of Blokhuis, Seress, and Wilbrink, a set whose pairwise joins are passant is an exterior set of the conic [3, pp. 143,146]; its vertices may have either internal or exterior point type. Every conic-filling arc is such an exterior set.

There is already a structural reduction at $q = 13$. An exterior point lies on only $(q-1)/2 = 6$ passants, fewer than the seven distinct chords through a vertex of an eight-arc. Hence every vertex would be internal, and $q = 13 = 2k - 3$ puts Lemma 4.1 in force: the seven chords at a vertex exhaust its complete passant pencil. Its characteristic vector would therefore be a weight-eight word of the code in Theorem 4.2, which is impossible. This gives the conceptual $q = 13$ exclusion needed for conic filling.

The stronger assertion that every pairwise-passant arc has at most six points still needs the orbit search. Join two off-conic points when their line is passant. The desired eight-arc would be an eight-clique in this graph, with collinear triples forbidden.

The finite certificate is a root-edge orbit DAG. Its exact data are:

q	passant edges	edge orbits	DAG nodes	six-arc orbits
13	7098	10	604	2
17	20808	13	4442	22
19	32490	15	11260	94.

For one representative of each edge orbit, every node lists all extension orbits under its stabilizer. Each terminal node assigns to every off-conic point either membership, a selected vertex joined to it nonpassantly, or a selected pair collinear with it. Rooted transition masses and the global identity prove completeness. More precisely, if $m_{B,k}$ is the sum of root-stabilizer orbit masses at level k and $e(S)$ is the number of valid extensions of a node S, the verifier checks

$$\sum_{S \in D_{B,k}} |\operatorname{Orb}_B(S)| e(S) = (k-1) m_{B,k+1}$$

for every root and level, together with

$$\sum_B |\operatorname{Orb}(B)| m_{B,k} = \binom{k}{2} M_k$$

globally. Hence no labelled arc is omitted. There is no node of size seven. A separately specified increasing-index backtracker reproduces every labelled level without group actions or canonical keys, and the earlier C++ search gives a third exact replay. The certificate includes a six-point witness in each field, proving sharpness. □

Remark 5.2 (The terminal bound six is not uniform in q). The last assertion of Theorem 5.1 is confined to $q \in \{13, 17, 19\}$ because it fails at $q = 23$. With $\mathcal{C} : XZ = Y^2$ in $\mathrm{PG}(2, 23)$, the eight points

$$(1{:}0{:}1),\ (1{:}1{:}2),\ (1{:}2{:}15),\ (1{:}5{:}15),$$
$$(1{:}7{:}20),\ (1{:}12{:}18),\ (1{:}13{:}11),\ (1{:}16{:}11)$$

form an arc off $\mathcal{C}$ all of whose twenty-eight joins are passant. The check is direct: writing $Q = Y^2 - XZ$ with polarization B, the join of two off-conic points P, R is passant exactly when $B(P, R)^2 - 4Q(P)Q(R)$ is a nonzero non-square, which holds for each of the twenty-eight pairs, and none of the 56 triples is collinear. Six of the vertices are internal and two external, so the configuration does not have a single point type.

The bound six is therefore an order-specific phenomenon rather than a uniform one. The counting, character-sum, and clique estimates tried here enter the order only as a parameter and do not distinguish $q = 19$ from $q = 23$. What closes $q = 13$ instead is the arithmetic coincidence $k - 1 = (q + 1)/2$ of Lemma 4.1, which holds at that order alone; at $q = 17$ and $q = 19$ the passant pencils are unsaturated by two and by three lines, external vertices become admissible, and the tangent product stops being a function of the conic alone, which is why those two orders are closed by the terminal search.

In coding terms, let C be a projective $[k, k-3, 4]_q$ MDS code with $4 \le k \le 8$. If its projective distance-three syndrome locus is a nonsingular conic, then, up to monomial equivalence, C is the $[4, 1, 4]_5$ frame code or the $[6, 3, 4]_{11}$ Clebsch code. Both codes have the stated syndrome locus.

Thus, for each fixed arc size, only finitely many field orders can admit conic filling; the classification is complete through eight points.

6 Trust and exact replay

Every claim is assigned one of exactly five proof modes. A *human structural proof* is a complete argument in this paper or the geometric paper. A *published theorem* is an external result used with its hypotheses matched. A *Lean theorem* is an exact named declaration in the pinned formal checkout. A *finite certificate* is a finite proof object checked by a small verifier, including a coverage or mass identity. A *trusted execution* reconstructs and exhausts its stated domain by an exact program but has no separately checked proof object.

Table 2 splits mixed theorem statements into atomic claims so that no row hides a change of trust. In particular, the association-algebra span argument and four-anchor rigidity are human proofs, not Lean claims. Conversely, the weight-ten exclusions and the terminal three-field bound remain finite certificates; the failed spectral and first-order linear-programming bounds are not cited as proofs. The exhaustive $q = 11$ conic-distance audit carries no row at all: rigidity makes it redundant, and it survives only as reported computation inside the proof of Theorem 2.4.

The A_5 point-orbit decomposition and complete syndrome oracle in Paper I are human structural proofs as well: the former closes by eigenspaces, stabilizers, and orbit–stabilizer, and the latter by the arc–coset dictionary and three chord-incidence equations. The q11 tables replayed here prove the sharper fifteen-class census and gap statements and provide independent corroboration; they are not premises of those main-paper arguments.

Atomic claim	Proof mode	Exact boundary
Chord defect, field window, and $q = 11$ rigidity	human structural proof	Theorems 1.1–1.3
$q = 9$ Sylvester identification and clique number five	published theorem	The pinpoint citations in Lemma 3.1
Chord-moment, $q = 9$, and small-k formal terminals	Lean theorem	The named declarations and axiom audit in the pinned gate
Fifteen $q = 11$ classes and their uncovered sizes	finite certificate	Orbit masses, concurrence counts, and chord defect
Low-degree rigidity	finite certificate	Fourteen cubic minors and the symbolic Clebsch kernel
$q = 13$ weight-eight exclusion	human structural proof	Segre reduction and the five-row unique closure
$q = 13$ weight-ten exclusions	finite certificate	Both exhaustive XOR-disjointness domains
$q = 13$ minimum-word classification and reconstruction	trusted execution	All four generated orbits and every pair/triple profile
$q = 13$ orbit spans and automorphism group	human structural proof	The mod-two elliptic intersection algebra and the four resolving anchors
$q = 11, 13$ seven-arc exclusions	finite certificate	Three orbit masses and three quadratic minors
Maximum passant-arc size at $q = 13, 17, 19$	finite certificate	Root-edge DAGs, transition masses, and terminal blockers
Independent labelled and legacy terminal replays	trusted execution	Increasing-index Python and discriminant/C++ searches

Table 2: The five-mode claim map. Supporting replays do not change a structural or certificate proof into a trusted-execution claim.

The machine-readable version of this ledger is the file

`verification/computational_companion_trust.json`.

Its paths are relative to the present directory. Recorded outputs and source hashes are in `verification/checker_outputs.json`. The terminal $k = 8$ certificate is checked by `verification/terminal_orbit_dag.py`.

The program `verification/conic_filling_verify.py` is instead a legacy independent replay. The clean companion entry point is

`verification/verify_computational_companion.py`.

Passing `--run` also executes every replay in Table 3.

Result	Exhaustive or independent executable check
Fifteen classes, gap, low-degree rigidity, and seven-arc leaves	`verification/build_finite_census_certificates.py` and its `--audit` mode
$q = 9$ exclusion and uniqueness of $q = 11$	`check_small_q_uniqueness.py`
$q = 13$ weight-ten profiles	`verification/q13_weight10_profiles.py --check` and `verification/q13_weight10_independent.py`
$q = 13$ code, minimum layer, structural identities, and old audits	`check_q13_tangent_code.py`
Root-edge DAGs and independent labelled replay	`verification/terminal_orbit_dag.py` and `verification/terminal_orbit_dag_replay.py --check`
Legacy passant-arc replay for $q = 13, 17, 19$	`verification/conic_filling_verify.py`

Table 3: Load-bearing certificate checks and trusted exact replays. Every finite command is deterministic and uses no random seed.

No exhaustive classification in this paper is certified by Lean. The gate contains, among others, the declarations

- `RelativeConicArcs.ClebschChordDefect.chordDefect_identity_of_moments`;
- `RelativeConicArcs.Q9Sylvester.distanceTwo_clique_number_five`; and
- `RelativeConicArcs.SmallKGeometricBridge.sevenArc_primePower_conic_card_spectra`.

Its tracked axiom audit reports the ordinary logical axioms and the two explicit Dye assumptions used by the rigidity implication. None of these declarations asserts the q11 census, q13 distance or minimum-word classification, or the q13/q17/q19 terminal DAG theorem. Those retain the finite modes stated in Table 2.

7 Conclusion

The finite census exposes two strengthenings that the geometric proof does not need: degree-three containment already characterizes the Clebsch class, and uncovered-set size has a four-point gap above it. Across fields, the Sylvester obstruction makes $q = 11$ unique for conic-filling six-arcs. Together with the terminal searches, this leaves exactly two conic-filling projective codes through length eight. At the first terminal field, the passant-code argument also identifies the exact binary distance and shows that the complete minimum layer reconstructs its passant geometry.

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